



DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
THIRD SESSIONAL

SUBJECT: (CE-618) NETWORK & INFORMATION SECURITY

Examination : B.Tech Semester VI
Date : 23-03-2023
Time : 2:30 pm to 3:45 pm

Seat No. : CF137
Day : Thursday
Max. Marks : 36

INSTRUCTIONS:

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed.

- (a) Apply shift row operation on the given state obtained after sub byte step in AES. [12]

CO4 A

$$S/\begin{bmatrix} C9 & FE & 30 & 63 \\ 63 & F2 & F2 & 26 \\ C9 & 7D & C9 & D4 \\ 82 & 63 & FA & D4 \end{bmatrix}$$

- (b) Compute the value in hex of round constant RC_9 with respect to polynomial $x^8 + x^4 + x^3 + x + 1$. [2]
- (c) Compare transport mode and tunnel mode in IP Sec protocol. [2]
- (d) Modify basic digital signature scheme such that it can also achieve confidentiality. Explain your answer. [2]
- (e) In which cases certificate revocation is required in X.509? Explain any two in brief. [2]
- (f) Distinguish between MDC and MAC. [2]

Q.2 Attempt Any TWO from the following questions.

- (a) In AES round, in mix column step, calculate $S_3(1,1)$ if we perform $S_3 = C \times S_2$. Assume, first row of matrix C is [2 3 1 1] and first column of matrix S_2 is [63 F2 7D D4]. Multiply first row by the first column and obtain the result. Consider polynomial $x^8 + x^4 + x^3 + x + 1$ for modulo operation. [6]
- (b) Consider sub byte computation in AES. Assume that multiplicative inverse output $S(r,c)^{-1}$ found is $(x^6 + x^5 + x^3 + 1)$. X is given below. Represent the polynomial $(x^6 + x^5 + x^3 + 1)$ as binary, transform it as 8x1 matrix (lsb to msb) and evaluate $X \cdot S(r,c)^{-1}$. [6]

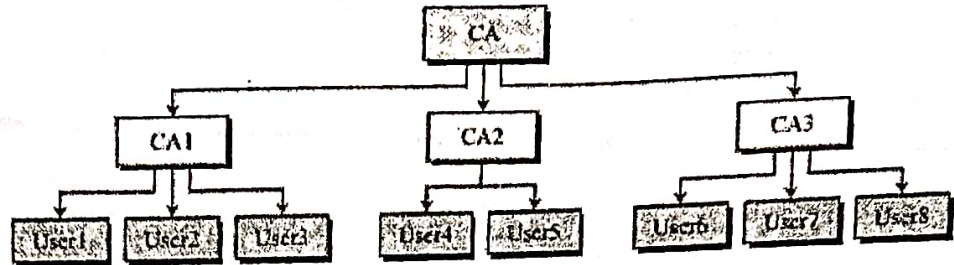
1100 0011

$$X = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

- (c) Construct a pseudo code for key expansion in AES-128. Give brief explanation of operations. [6]

Q.3 Attempt the following questions.

- CO2 R** (a) Explain Needham-Schroeder key sharing protocol with neat and clean diagram.
- CO3 A** (b) Explain Certification authority method of public key distribution in detail and apply that to solve the following question: Show how user1, knowing only the public key of the CA (the root), can obtain a verified copy of User8's public key? Use the following figure to answer the question.



OR

Q.3 Attempt the following questions.

- CO2 R** (a) Assume Alice (the client) wants to create a shared secret key with Bob (the server) using AS and TGS. Explain all 6 steps in detail with example.
- CO3 A** (b) Initial values of Buffers are as below.
 $A=0X7$ $B=0XA$ $C=0XC$ $D=0X4$ $E=0X9$ $F=0Xa$ $G=0Xf$ $H=0X6$. If these initial values are given to round 0 of SHA-512 then what will be the value of buffer A B C D E F G H after round-0? ($W0=0X7$ $K0=0X9$ $Rotate(X)$ =circular rotation in left by 1 bits. Use addition mod 2^4).